

TerraMind

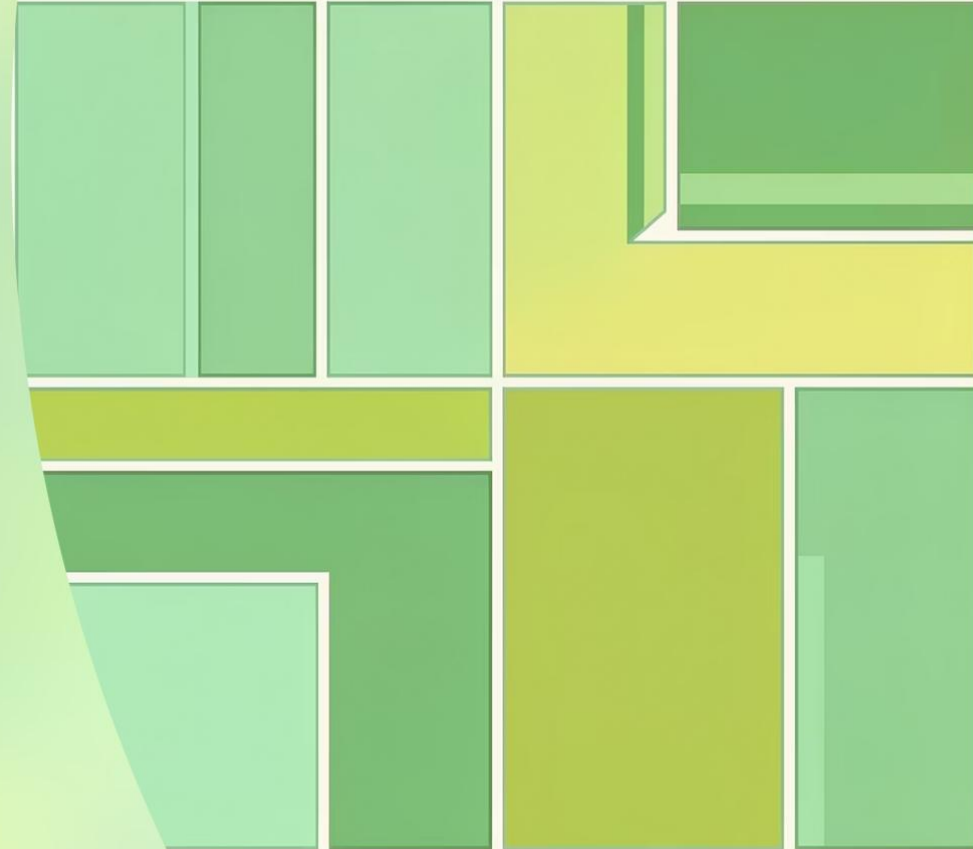
Give Every Field a Memory

A temporal knowledge-graph digital twin for farmland –
powered by **Cognee**

LONGITUDINAL AGRONOMY & SOIL MEMORY

THE HANGOVER HACKATHON

💡 Team MetaData





Farming Runs on Multi-Year Memory, But Records Don't.

System Fault: Data Fragmentation

T-2 YEARS



Scattered Inputs

Critical chemical applications & soil interventions are locked in unstructured formats: handwritten ledgers, disparate PDFs, and isolated supplier receipts.

T-1 YEARS



Lost Environmental Context

Weather APIs track the sky, but biological shifts in the soil memory—caused by historical droughts or floods—remain untracked and entirely disconnected.

PRESENT DAY



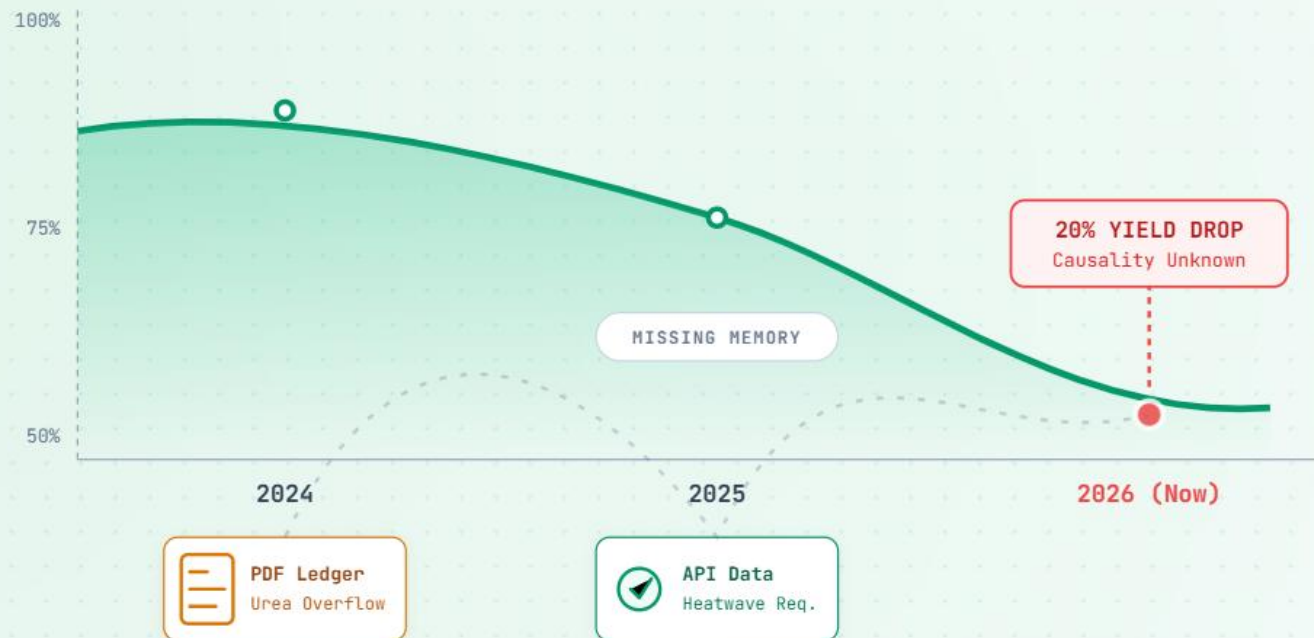
Unexplained Anomalies

Crop failures are treated as isolated events. Agronomists rely on gut instinct because no system connects multi-hop temporal data to diagnose the true root cause.

Yield vs. Unstructured Data Silos

Visualizing the multi-year temporal disconnect

SIMULATED RUN // PLOT B



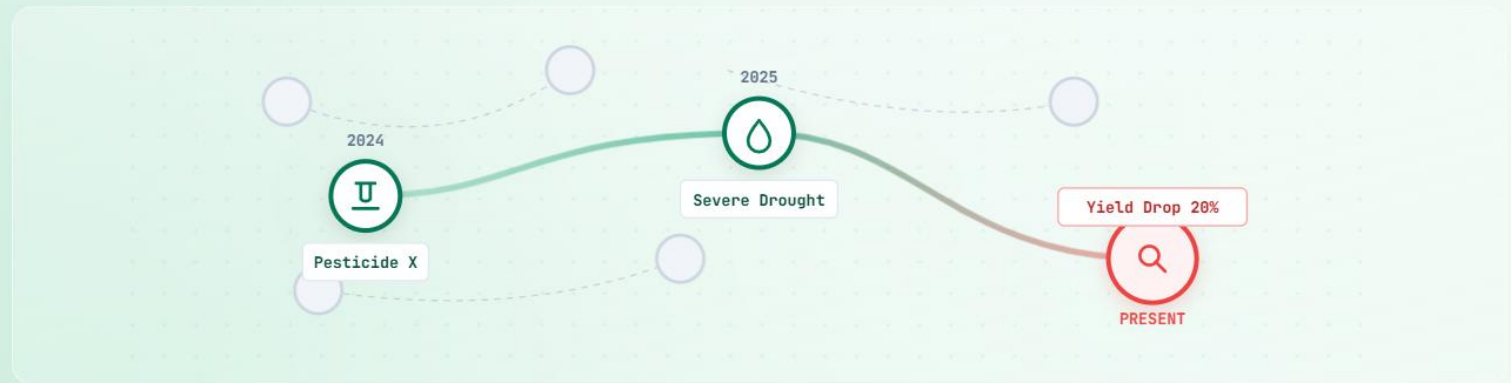
A Digital Twin Memory Graph — For Every Plot.

TerraMind ingests historical weather data, field notes, and chemical logs for a plot of land, then performs multi-hop graph traversal to surface the connected chain of events behind an anomaly.

Every answer ships with a visible evidence trail and a confidence label, never a bare claim.

THE MEMORY GRAPH

Query → "Why did Field B's yield drop by 20%?"



STATISTICAL ASSOCIATION // 3 LINKED TEMPORAL RECORDS

01 INGEST

Weather, Field Notes & Logs

Historical disparate unstructured data is continuously ingested and standardized for every plot.

02 TRAVERSE

Multi-Hop Graph Reasoning

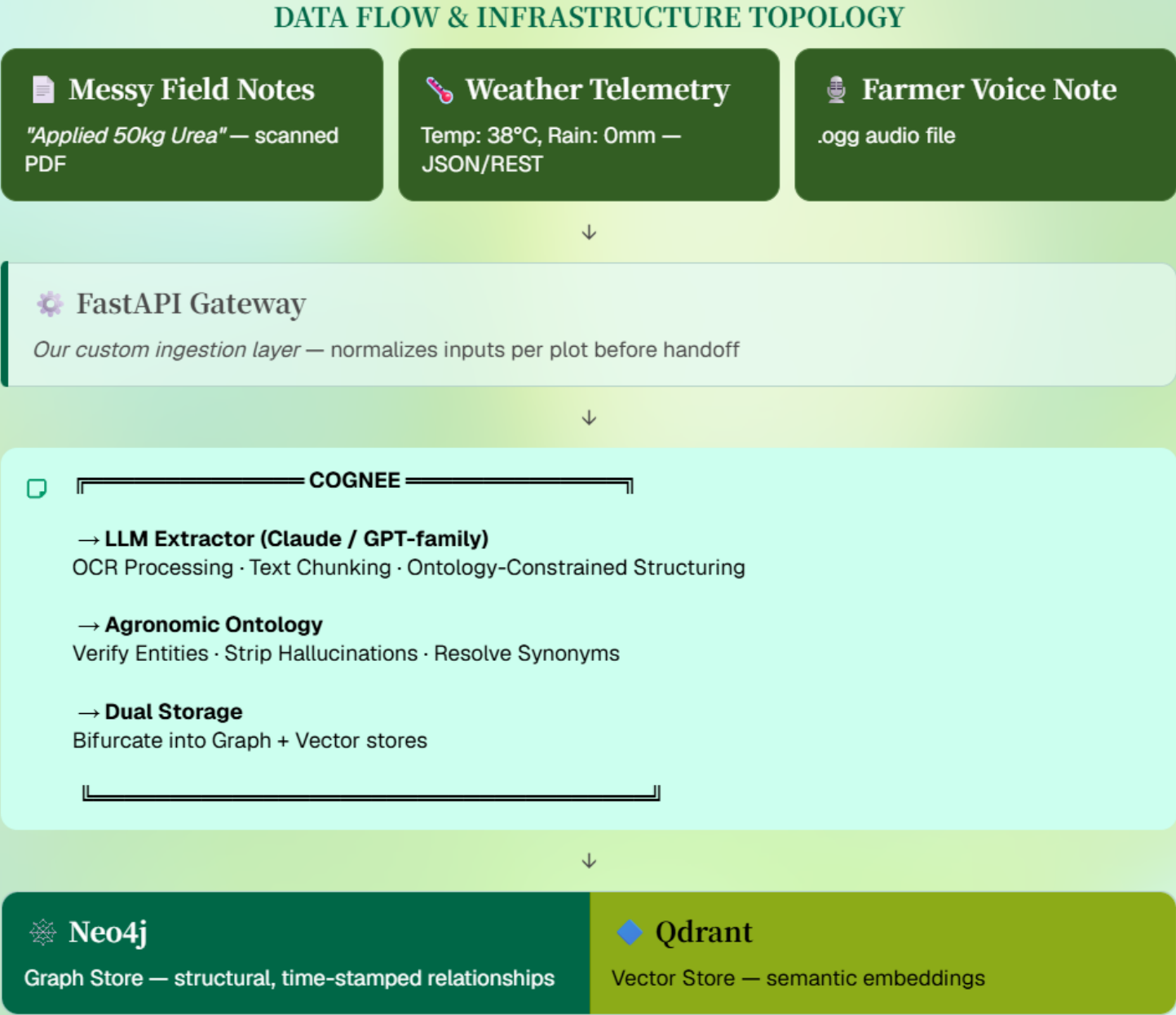
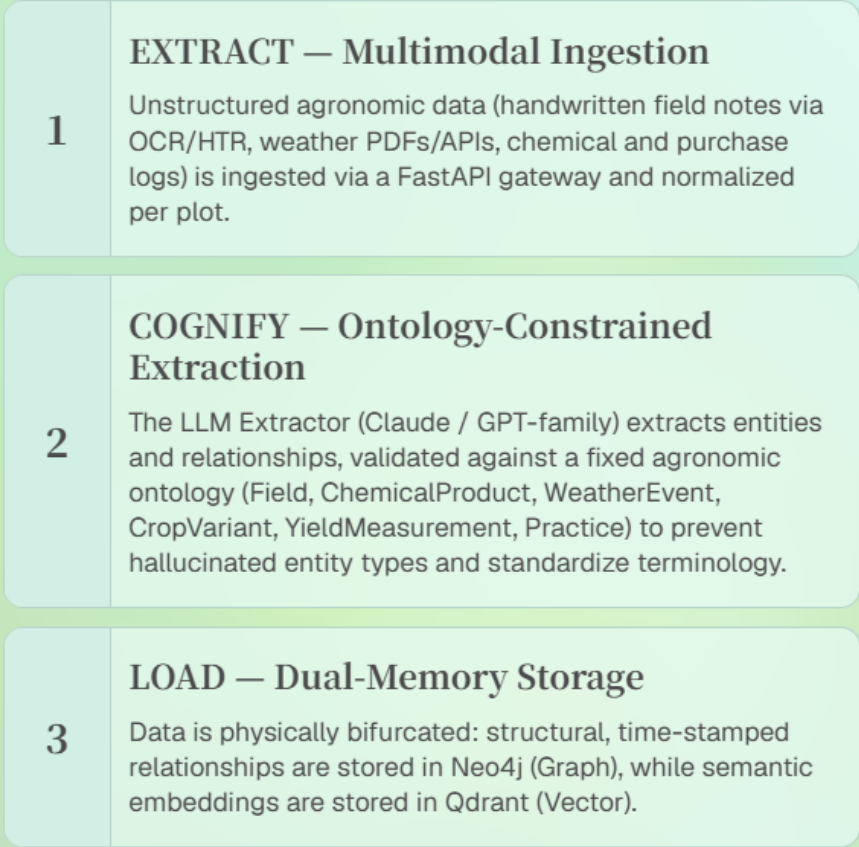
The engine performs temporal graph traversal across years to surface hidden causal event chains.

03 EXPLAIN

Evidence Trail & Labeling

Results are never black-box. Every answer ships with a visible evidence graph and confidence score.

The Extract, Cognify, Load (ECL) Pipeline



Tech Stack



⬆ INGESTION

FastAPI Gateway + OCR

Handwriting & document parsing entry point.



</> MEMORY ENGINE

Cognee Framework

Extract → Cognify → Load pipeline.



⚙ LLM

Claude / GPT-family

Entity extraction & answer synthesis.



🔍 VECTOR STORE

Qdrant

Semantic similarity & embedding search.



🔗 GRAPH STORE

Neo4j

Time-stamped entities & structural relationships.



⚡ CACHE & QUEUE

Redis + Celery

Semantic query cache & ingestion backpressure.



🔒 AUTH & SECURITY

OAuth 2.0 / JWT

Per-farm tenant scoping and API security.



</> FRONTEND

React / Next.js

Chat, timeline & complex graph visualization.

📦 DEPLOYMENT

Docker Compose (demo) → containerized cloud, per-tenant isolation (scale)

📊 OBSERVABILITY

Structured logs, metrics, tracing

Trust by Design — Evidence Trails & Confidence Labels

Confidence Tiers

✓ DOCUMENTED FACT

Directly stated in a source record — highest confidence, fully traceable to a primary document (e.g., specific chemical ledger).

≡ STATISTICAL ASSOCIATION

A graph co-occurrence across linked records — no confirmed mechanism, amber-flagged for agronomist review.

⌚ UNCONFIRMED HYPOTHESIS

A plausible but weakly-supported connection — shown only on explicit request, gray/outline UI treatment.

Sample Answer Card

PRODUCT MOCKUP

🔍 QUERY

"Why did Field B's yield drop by 20%?"

⚡ AI CONCLUSION

The yield drop is statistically associated with a legacy chemical application compounded by recent weather stress.

≡ EVIDENCE TRAIL



PDF: Fertilizer Log

2024

Pesticide X applied



API: Weather Data

2026

Severe drought recorded



DERIVED: Graph Interaction

MULTI-HOP

Pesticide X × drought — 3 linked records



Every claim traces to a specific graph edge. No traceable edge, no claim shown.

>_ GROUND-TRUTH EVAL HARNESS

15–20 agronomist-verified historical cases scored on **correct cause** and **correct confidence label** — run as a regression gate on every model or prompt change to ensure answer quality never regresses.

Why TerraMind Wins

 NOVEL DOMAIN

AgriTech + temporal graph-RAG is a genuinely underused, high-impact combination at AI hackathons. It breaks the mold of generic enterprise SaaS bots.

 REAL TECHNICAL FIT

Cognee’s multi-hop and temporal reasoning are purpose-built for exactly this problem. This is a profound architectural integration, not a generic RAG wrapper.

 TRUST-FIRST

Evidence trails and confidence labels proactively answer the hardest question before a judge even asks it: "How do you know it's not just correlation or hallucination?"

 GROUNDED SCOPE

A **diagnostic aid**, not a prescriptive oracle. By focusing on root-cause analysis, the product is lower-risk, defensible, and genuinely useful from day one.

MVP
Single plot, reviewed notes, UI

1 PILOT
Multi-plot, mobile capture, human-loop

2 SCALE
Multi-tenant hardening, load testing

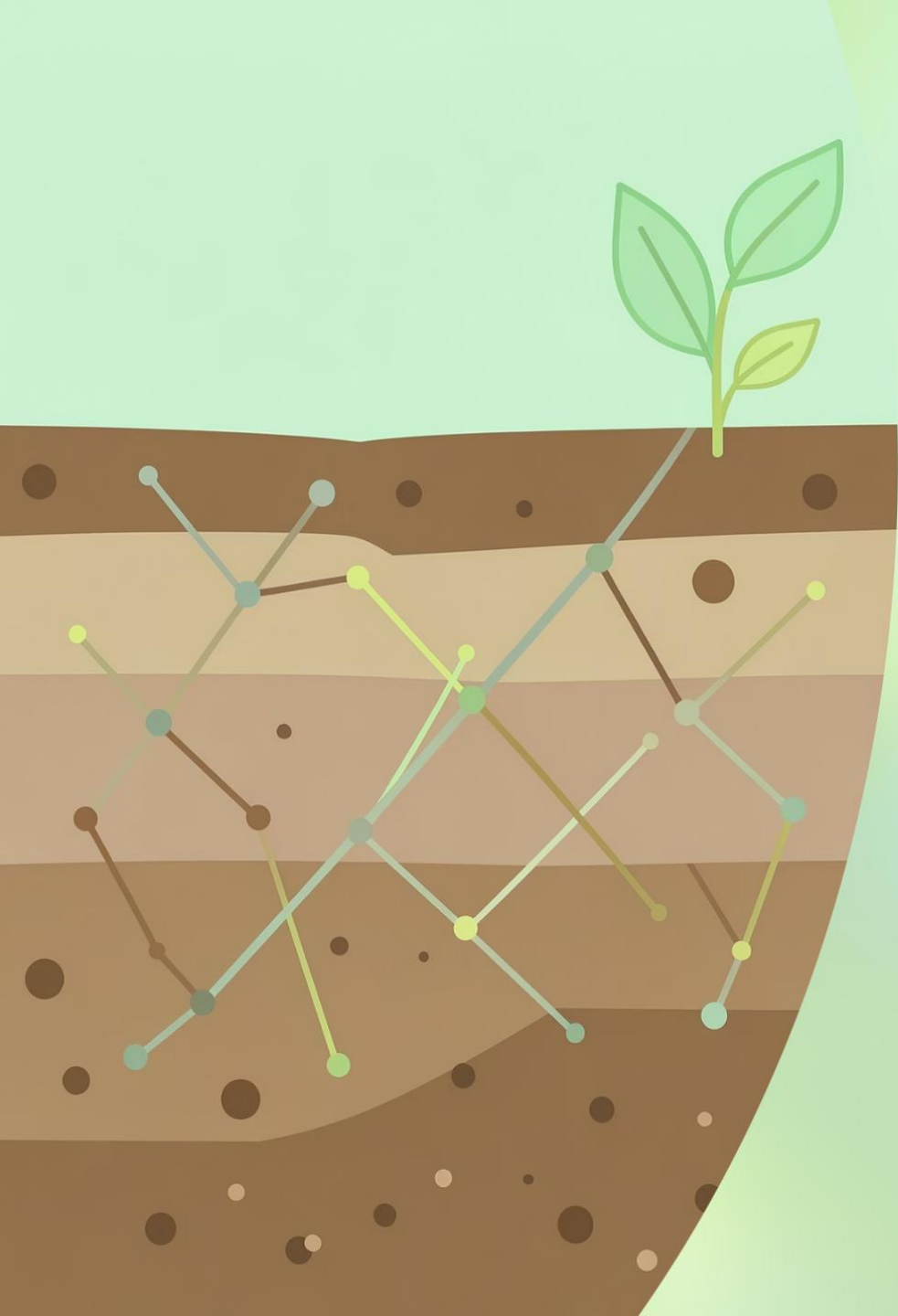
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TerraMind — Give every field a memory.

 Team: MetaData

OUR ASK

The Hangover Hackathon — We are seeking feedback, mentorship, and a pilot farm partner to validate this temporal evidence-trail approach in the physical field.



Thank You

Give every field a memory.

TerraMind – Longitudinal Agronomy & Soil Memory | The Hangover
Hackathon | Team MetaData

Questions & feedback welcome.